copulaendog: instrument-free copula corrections for endogenous regressors in Stata and Python

Ported from the R implementations of
Rouven E. Haschka and Ashwin Malshe

Abstract

`copulaendog` corrects endogenous regressors in a linear model when no instrument is available. The dependence between the endogenous regressor and the structural error is modelled with a Gaussian copula, and the resulting control function is added to the regression. Five cross-sectional estimators are implemented behind one interface: Park and Gupta (2012), the two-stage copula generated regressor approach of Yang, Qian and Xie (2025), the between-regressor correlation variant of Haschka (2025a), the nonlinear-transformation approach of Breitung, Mayer and Wied (2024), and the adjusted estimator of Liengaard et al. (2025). Seven marginal CDF estimators are available for the copula transformation, inference is by pairs bootstrap, and a `validity` report walks the identification requirements the literature states. This document describes the Stata command `copulaendog` and the companion Python package `copulaendog`, both ported from the R reference implementation of Haschka and from the `endogCopula` package of Malshe.

1 Credit

This software is a port. The econometrics, the design of the interface, and the reference code are the work of two authors, and neither the Stata command nor the Python package would exist without them:

- **Rouven E. Haschka**, *Copula-based endogeneity corrections in R*, `https://github.com/HashtagHaschka/Copula-based-endogeneity-corrections` ORCID `https://orcid.org/0000-0002-2916-9745`. This is the reference implementation. Every estimator, every CDF estimator, the bootstrap, the diagnostics and the validity report are ported from it.
- **Ashwin Malshe**, *endogCopula: Endogeneity Corrections via Gaussian Copulas*, `https://github.com/ashgreat/endogCopula`. The packaged R version of the same estimators, which shaped how the port is organised into modules and what the documented interface looks like.

If this code contributes to published work, please cite both of those, and the paper behind whichever estimator was used. The list is in Section 7.

2 The model

Let y be the outcome, P a vector of endogenous regressors and W a vector of exogenous ones. The structural model is

$$y_i \;=\; \mu + P_i'\alpha + W_i'\beta + \xi_i, \qquad \operatorname{corr}(P_i, \xi_i) \neq 0, \tag{1}$$

so OLS is biased. With an instrument one would run two-stage least squares. The copula approach replaces the instrument with a distributional assumption: the dependence between P and ξ is a Gaussian copula. Writing $P_k^* = \Phi^{-1}(F_k(P_k))$ for the normal score of the kth endogenous regressor and $\xi^* = \xi/\sigma$, the assumption is that (P^*, ξ^*) is jointly normal. Then

$$E[\xi_i \mid P_i] \;=\; \sigma \sum_k \rho_k P_{ik}^*, \tag{2}$$

and adding $\hat{P}^*$ to (1) as a control function removes the bias. The estimator is ordinary least squares on the augmented regression

$$y_i \;=\; \mu + P_i'\alpha + W_i'\beta + C_i'\gamma + u_i. \tag{3}$$

What identifies the model is *non-normality* of P. If P is normal then $\Phi^{-1}(F(P))$ is an affine function of P itself, C is collinear with the regressors, and (3) has no solution. This is not a technicality to be checked afterwards: it is the entire source of identification, and it is why the `validity` report leads with the shape of P.

In a finite sample the failure is quieter than that, and worth recognising. $\hat{F}$ is not the true normal CDF, so a near-normal P leaves a design that is nearly rather than exactly singular: the model estimates, and the estimate is badly wrong. On $n = 400$ draws with a standard normal P and a true α of 2.0, the correction returns 2.54 with a standard error of 0.51 against an OLS standard error of 0.03 — an ICON above 15. Nothing errors out. The diagnostics are what catch it, which is why they are printed rather than merely stored.

The five estimators differ in how C is built.

2.1 The estimators

PG, Park and Gupta (2012). The original. $C_k = \Phi^{-1}(\hat{F}_k(P_k))$, added directly. Assumes $\operatorname{corr}(W, P^*) = 0$. This assumption is what motivated everything that followed, and it is testable: `validity` reports it.

2sCOPE, Yang, Qian and Xie (2025). Transforms the exogenous regressors too, and residualises:

$$C_k = P_k^* - \delta_k' W^*, \tag{4}$$

where $W^* = \Phi^{-1}(\hat{F}(W))$ and δ_k comes from a first-stage regression of P_k^* on W^* alone — never on the other endogenous regressors. The correction is then orthogonal to W by construction, so the uncorrelatedness assumption is not needed. The first stage carries an intercept.

IMA, Haschka (2025a). The same construction without an intercept in the first stage, which is what the paper's derivation of ρ requires: the slope of P^* on W^* is then the Pearson correlation between the two. In practice the two estimators are very close; the gap widens the further the transformed exogenous regressors sit from mean zero.

BMW, Breitung, Mayer and Wied (2024). Reverses the order of the two operations. First regress P on W by OLS on the *raw* variables and keep the residual $\hat{e}$; then take its rank transform, $C = \Phi^{-1}(\text{rank}(\hat{e})/(n+1))$. Same two ingredients, opposite order, and the two are not reparameterisations of one another: their abstract puts the difference plainly, that 2sCOPE "calculate[s] residuals from a regression of the transformed regressors, thereby assuming a particular nonlinear dependence." Two consequences show up in the output. The identification requirement now falls on $\hat{e}$ rather than on P, so `validity` tests $\hat{e}$; and their Corollary 3.2 makes the textbook t statistic valid for testing $\rho = 0$, so a Durbin–Hausman–Wu test is reported next to the bootstrap one.

JAMS, Liengaard et al. (2025). Lets the copula structure differ across the categories of the discrete exogenous regressors. With d_P endogenous and d_W continuous exogenous regressors,

$$C(P_i, W_i) = \left(C(P_i)' \;\; C(W_i)'\right) \Sigma^{-1} \begin{bmatrix} I_{d_P} \\ 0 \end{bmatrix}, \tag{5}$$

with Σ the variance–covariance matrix of $(C(P), C(W))$ — the covariance matrix, not the correlation matrix; the two give different linear combinations and therefore different estimates. Conditioning on a set of categorical controls estimates Σ and the marginal CDFs inside each joint category, so every copula column is zero outside its own cell.

2.2 Estimating the marginal CDF

The literature disagrees on how $\hat{F}$ should be estimated, so all of the proposals are implemented and the choice is free (except in BMW).

Option	Source	What it is
`kde.silverman`	Park & Gupta (2012)	integrated Epanechnikov kernel density, Silverman bandwidth
`kde.cv`	Li, Li & Racine (2017)	the same, cross-validated bandwidth
`kde.plugin`	Polansky & Baker (2000)	Gaussian kernel CDF, plug-in bandwidth
`ecdf.fixed`	Becker et al. (2022)	empirical CDF with replaced boundary
`ecdf.adj`	Liengaard et al. (2025)	adjusted ECDF
`rank.n`	Qian, Koschmann & Xie (2025)	rank/n with a top correction
`rank.n1`	Breitung et al. (2024)	rank/(n+1)

Defaults follow each estimator's own paper: `kde.silverman` for PG, `rank.n` for 2sCOPE and IMA, `ecdf.adj` for JAMS, `rank.n1` for BMW. BMW is the one place where the choice is not free, because its Proposition 3.1 is derived for rank/(n+1).

`kde.cv` is available in Python but not in Stata: it needs an $O(n^2)$ cross-validation inside every bootstrap replicate.

2.3 Inference

C is a generated regressor, so the textbook OLS standard errors from (3) are wrong for the structural coefficients. Inference is by a plain pairs bootstrap: draw row indices with replacement, recompute the copula terms on the resample, refit. This is the procedure used in every one of the underlying papers. Resamples that lose a factor level, or that leave a JAMS category too thin to estimate its own Σ, are rejected and redrawn; if that happens often the command says so, because the standard errors are then conditional on the draws that survived.

Each fit also carries the uncorrected OLS estimate from the *same* resamples. The ratio of the two standard errors is ICON, the standard error inflation of Qian, Koschmann and Xie (2025). Values above 6 flag weak identification or a misspecified dependence model.

3 The Stata command

3.1 Syntax

```
copulaendog depvar endogvars [if] [in] [, exog(varlist) method(name)
    cdf(name) ties(name) nboots(#) seed(#) conditional(varlist)
    discrete(varlist) fsexclude(varlist) generate(stub) noconstant
    level(#) validity ]
```

Position decides. Variables listed after *depvar* are endogenous and each gets its own copula term; everything in `exog()` is exogenous and gets none. Endogenous regressors must be plain numeric variables, because a factor variable expands into several columns and has no single copula term; put categorical controls in `exog()`, where factor-variable notation is allowed.

Option	Default	Meaning
`exog()`	—	exogenous regressors, factor variables allowed
`method()`	`pg`	`pg`, `2scope`, `ima`, `bmw`, `jams`
`cdf()`	per estimator	marginal CDF estimator
`ties()`	`max`	`max` for the counting function, `average` for midranks
`nboots()`	199	bootstrap replicates
`seed()`	—	seed; without it the draws are not reproducible
`conditional()`	—	JAMS: what the copula structure may vary over
`discrete()`	—	JAMS: exogenous regressors to keep out of C
`fsexclude()`	—	exogenous regressors to hold out of the first stage
`generate()`	—	keep the copula terms as variables
`validity`	—	report the identification checks

3.2 A worked example

The data are simulated so that the assumptions can be checked against the truth. The endogenous regressor is $P = \exp(0.8v + 0.6w)$ with $\mathrm{corr}(v, \xi) = 0.6$, and

$$y = 1 + 2P + 1.5x - 0.5w + \xi.$$

Because P is a strictly monotone function of the normal variate $0.8v+0.6w$, the Gaussian copula between P and ξ holds exactly, while P itself is lognormal and therefore strongly non-normal — exactly the setting the estimators are built for. Note the $0.6w$ term: it makes $\mathrm{corr}(W, P^*) \neq 0$, so the Park and Gupta assumption fails by construction and the later estimators should do better.

```
. use simdata, clear
. regress y P x w
. copulaendog y P, exog(x w) method(pg)     nboots(499) seed(1) validity
. copulaendog y P, exog(x w) method(2scope) nboots(499) seed(1)
. copulaendog y P, exog(x w) method(bmw)    nboots(499) seed(1)
```

On $n = 2000$ draws from that design the estimates of α, whose true value is 2.000, are as follows. (These are from the Python port, which has been checked against the R reference to 10^{-12}; the file is `validation/vignette_data.csv`, so they can be reproduced in any of the three languages.)

	$\hat{\alpha}$	boot. SE	$\hat{\rho}$	max ICON
OLS, uncorrected	2.2066	0.0106	—	—
PG	2.0039	0.0320	0.669	0.91
2sCOPE	2.0136	0.0258	0.470	1.12
IMA	2.0136	0.0258	0.470	1.12
BMW	2.0109	0.0220	0.512	1.54
JAMS	2.0125	0.0254	0.470	1.13

Uncorrected OLS is off by 0.21, about twenty of its own standard errors. All five corrections land within 0.014 of the truth, and every ICON is close to one, so the correction costs almost nothing in precision here. The price of the correction is visible all the same: the standard error on $\hat{\alpha}$ triples under PG.

The `validity` report on the same fit rejects the uncorrelatedness assumption — correctly, since we built the data that way, with $R^2 = 0.349$, $F = 535.1$ and $p < 10^{-100}$ for the joint test — and points the user at the estimators that do not need it. It also confirms that P is non-normal enough to identify the model on both the Yang and the Becker criteria (skewness 10.0, AD 182.9, KS $p < 10^{-90}$), which is what makes this design work at all.

Two endogenous regressors, a factor control, and JAMS with the copula structure varying over the categories of `store`:

```
. copulaendog y price adspend, exog(feature) method(2scope) seed(1)
. copulaendog y price, exog(feature i.store) method(bmw) seed(1)
. copulaendog y price, exog(feature i.store) method(jams) ///
      conditional(store) seed(1)
```

Prediction excludes the copula terms, because they are endogeneity controls and not part of the causal model:

```
. predict yhat
. predict xi, residuals
```

3.3 Reading the validity report

The report keeps three sources apart rather than merging them into one verdict, because they do not fully agree.

1. **Non-normality** of P — or of the first-stage residuals under BMW. Two criteria are shown. "Yang ok" is the Kolmogorov–Smirnov criterion of Yang, Qian and Xie (2025), $p < .05$, which they prefer precisely because it is conservative. "Becker ok" is the C5.0 decision tree of Becker, Proksch and Ringle (2022, Fig. 8), which combines skewness with the Anderson–Darling and Cramér–von Mises statistics and depends on the sample size. Becker et al. report a correlation of .66 between the AD statistic and the copula term's t statistic against $-.03$ for KS, which is why both are shown.
2. **The uncorrelatedness assumption**, printed only for PG. Both the Holm-adjusted single correlations and a joint test are reported; the assumption is about the whole set, so the joint test is the one that matters.
3. **The structural error.** Becker et al. find that a non-normal error breaks consistency in models with an intercept; Yang et al. and Qian et al. permit one under the decomposition $\xi = U + V$. The residuals show ξ rather than U, so their shape settles neither question, and the report says so.
4. **ICON**, the standard error inflation relative to uncorrected OLS.

4 The Python package

The same five estimators, the same seven CDF estimators, the same bootstrap:

```
import pandas as pd
from copulaendog import copreg, sim_endog

df  = sim_endog(n=2000, seed=123)
fit = copreg("y ~ P | x + w", df, method="2scope", nboots=499, seed=1)

print(fit.summary())
print(fit.validity())

fit.params, fit.bse, fit.vcov      # coefficients and bootstrap covariance
fit.conf_int(0.95, kind="percentile")
fit.rho, fit.icon                  # endogeneity measure, SE inflation
fit.predict(newdata)               # structural model only
```

The formula follows the R interface: terms before the `|` are endogenous, terms after it are exogenous. Everything `patsy` accepts works — `np.log(price)`, `price:feat`, `I(x**2)`, `C(store)`, and `- 1` to drop the intercept.

Individual estimators can be called directly — `CopRegPG`, `CopReg2sCOPE`, `CopRegIMA`, `CopRegBMW`, `CopRegJAMS` — and the copula transformation itself is exposed as `copula_transform(x, cdf, ties)` for use in models this package does not cover.

5 What is not ported

Three estimators from the R implementation are deliberately out of scope here, and the R code should be used for them:

- **2sCOPE-np** (Hu, Qian and Xie 2025), which replaces the first stage with a nonparametric conditional CDF estimated by the kernel method of Li and Racine (2008). It is the only estimator in the family that tolerates discrete endogenous regressors.
- **PANEL** (Haschka 2022), a linear panel model with fixed effects estimated by maximum likelihood after a forward orthogonal deviations transformation.
- **BAYES** (Haschka 2025b), in which the marginal CDFs are unknown parameters drawn alongside the coefficients rather than plugged in.

6 Verification

The Python port is checked against the R reference implementation on a common dataset: across ten model specifications — all five estimators, four CDF estimators, an interaction, and JAMS with and without conditioning — all 75 coefficients and ρ values agree to within 10^{-12}, and the six closed-form CDF estimators agree to machine precision. The comparison script is in `validation/` and can be rerun.

The Stata port implements the identical algorithms in Mata and ships with a self-test do-file, `validation/stata_selftest.do`, which runs every estimator, compares the copula transformations against the same R reference, and prints the coefficients for comparison. Run it once in your own Stata installation before relying on the results.

7 References

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